PVsyst V8.1.8

PVsyst - Simulation report

Grid-Connected System

Project: soalr_trial

Variant: New simulation variant

Sheds, single array

System power: 78.3 kWp

SOLAR_TRIAL - India

Author

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VC0, Simulation date:
06/11/26 01:15
with V8.1.8

Project summary

Geographical Site
SOLAR_TRIAL
India

Situation
Latitude [redacted]
Longitude
Altitude 20 m
Time zone UTC+5.5

Project settings
Albedo 0.20

Weather data
SOLAR_TRIAL
Meteonorm 9.0 dll, Sat=100% - Synthetic

System summary

Grid-Connected System

Sheds, single array

Orientation #1
Fixed plane
Tilt/Azimuth 21 / -4 °

Near Shadings
According to strings : Fast (table)
Electrical effect 100 %

User's needs
Unlimited load (grid)

System information

PV Array
Nb. of modules 135 units
Pnom total 78.3 kWp

Inverters
Nb. of units 0.9 unit
Total power 72 kWac
Pnom ratio 1.09

Results summary

Simulation step Hourly

Produced Energy 102661 kWh/year | Specific production 1311 kWh/kWp/year | Perf. Ratio PR 73.32 %

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General parameters

Grid-Connected System — **Sheds, single array**

Simulation step
Hourly

Orientation #1

Fixed plane
Tilt/Azimuth 21 / -4 °

Sheds configuration
Nb. of sheds 9 units
Single array
Shading limit angle
Limit profile angle 43.9 °

Sizes
Sheds spacing 3.00 m
Sensitive width 2.28 m
GCR Shading 76.6 %
Top inactive band 0.02 m
Bottom inactive band 0.02 m

Models used
Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon
Free Horizon

Near Shadings
According to strings : Fast (table)
Electrical effect 100 %

User's needs
Unlimited load (grid)

PV Array Characteristics

PV module
Manufacturer Generic
Model HYPERSOL VSMDH.72.AAA.05
(Custom parameters definition)
HYPERSOL VSMDH.72.AAA.05.PAN
Unit Nom. Power 580 Wp
Number of PV modules 135 units
Nominal (STC) 78.3 kWp
Modules 9 units x 15 In series
At operating cond. (50°C)
Pmpp 72.9 kWp
U mpp 594 V
I mpp 123 A

Inverter
Manufacturer Generic
Model SH80CX
(Original PVsyst database)
Unit Nom. Power 80.0 kWac
Number of inverters 9 * MPPT 10% 0.9 unit
Total power 72.0 kWac
Operating voltage 180-1000 V
Max. power (=>30°C) 88.0 kWac
Pnom ratio (DC:AC) 1.09
No power sharing between MPPTs

Total PV power
Nominal (STC) 78 kWp
Total 135 modules
Module area 349 m²
Cell area 322 m²

Total inverter power
Total power 72 kWac
Nb. of inverters 1 unit
0.1 unused
Pnom ratio 1.09

Array losses

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses
Global array res. 79 mΩ
Loss Fraction 1.50 % at STC

Module Quality Loss
Loss Fraction -0.75 %

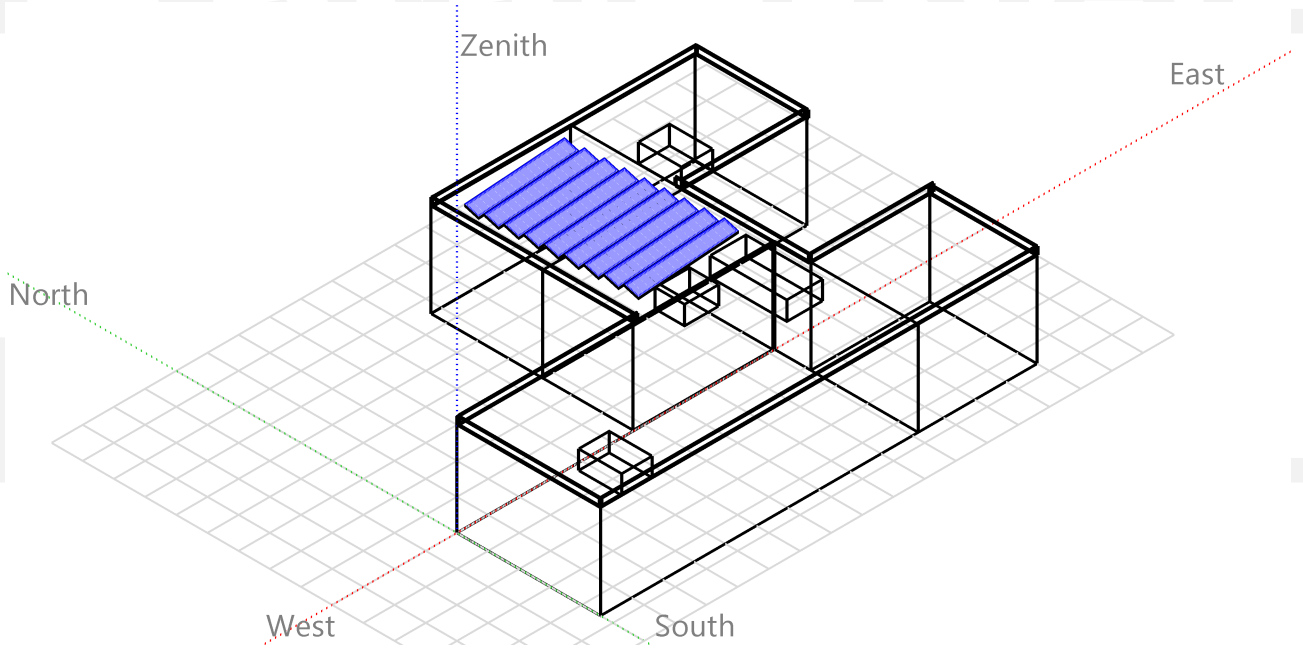
Module mismatch losses
Loss Fraction 2.00 % at MPP

IAM loss factor
Incidence effect (IAM): Fresnel smooth glass, n = 1.526

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.998	0.981	0.948	0.862	0.776	0.636	0.402	0.000

Near shadings parameter

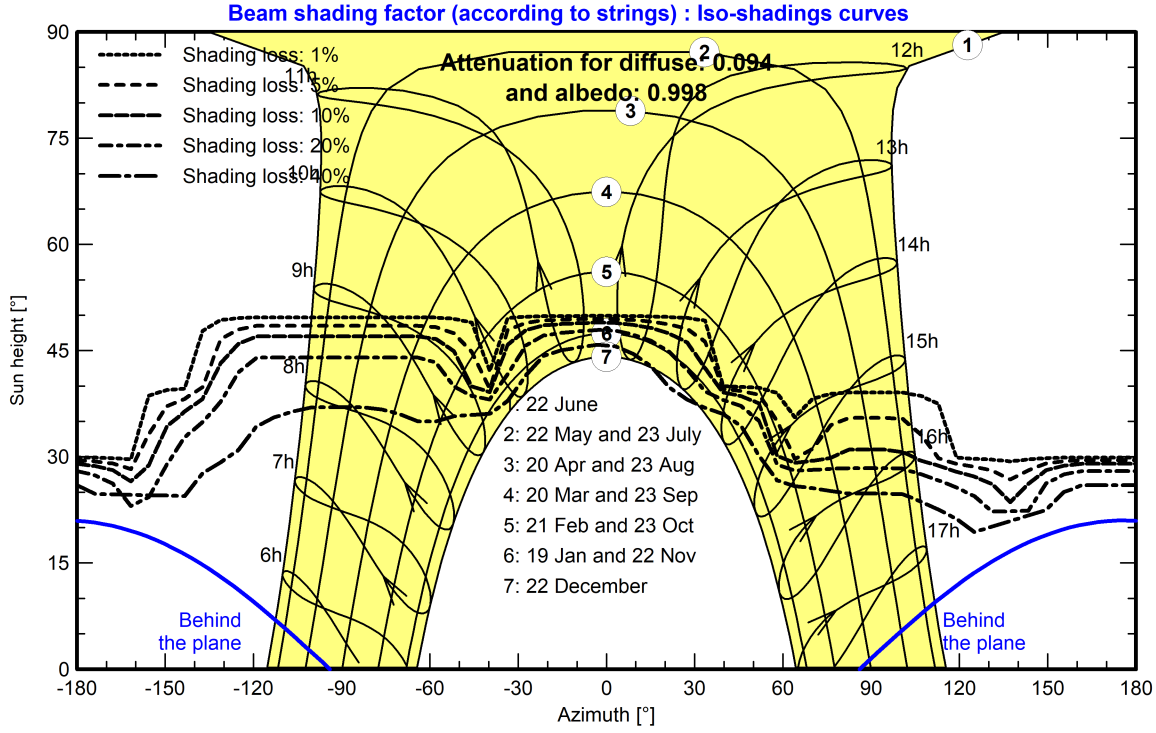
Perspective of the PV-field and surrounding shading scene



Iso-shadings diagram

Orientation #1 - Fixed plane, Tilts/azimuths: 21°/ -4°

Beam shading factor (according to strings) : Iso-shadings curves



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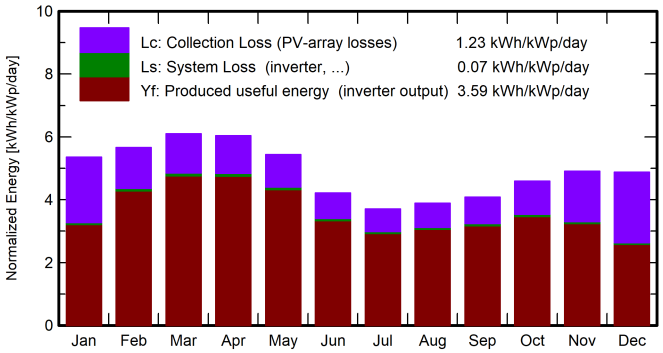
Main results

Simulation step
Hourly

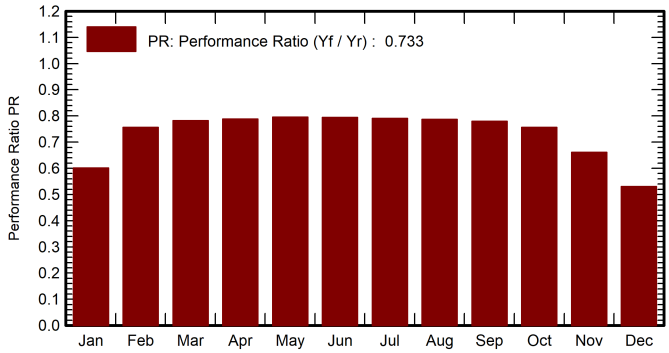
System Production

Produced Energy	102661 kWh/year	Specific production	1311 kWh/kWp/year
		Perf. Ratio PR	73.32 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_Grid kWh	PR ratio
January	131.4	51.59	17.47	165.9	152.6	7967	7808	0.601
February	134.9	56.01	22.13	158.5	149.3	9577	9390	0.757
March	174.1	78.94	27.26	189.0	178.9	11798	11575	0.782
April	180.2	88.51	30.41	181.0	170.9	11382	11164	0.788
May	178.3	95.52	31.20	168.7	158.4	10710	10503	0.795
June	136.3	89.20	30.80	126.4	116.7	8026	7858	0.794
July	122.8	82.90	29.63	115.0	105.4	7282	7120	0.791
August	123.7	79.14	29.26	120.6	111.4	7600	7435	0.787
September	119.2	77.61	28.60	122.4	113.0	7639	7472	0.780
October	127.2	64.00	27.20	142.3	132.8	8605	8431	0.757
November	120.4	56.32	23.50	147.2	135.4	7781	7627	0.662
December	118.6	51.34	18.77	151.1	136.6	6415	6278	0.531
Year	1667.3	871.08	26.37	1788.2	1661.2	104781	102661	0.733

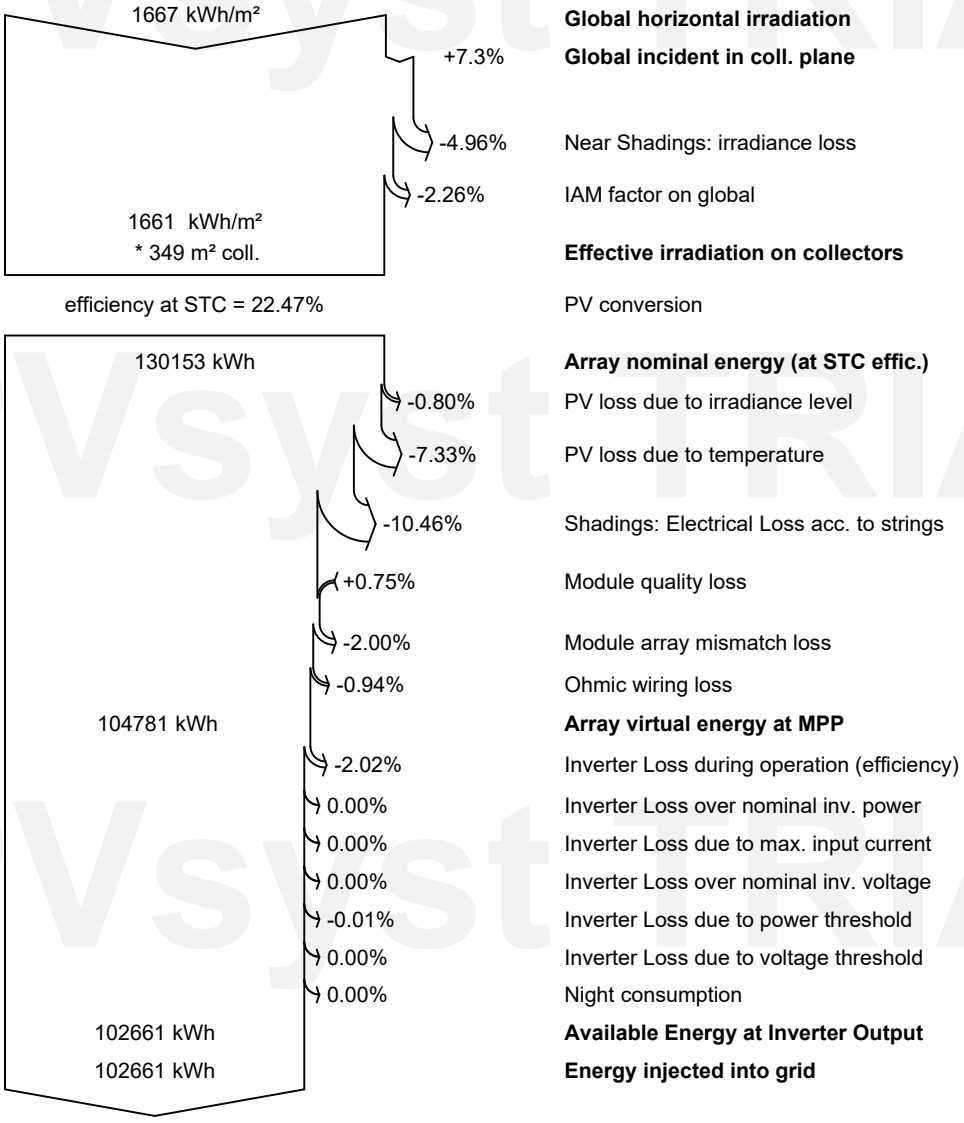
Legends

GlobHor	Global horizontal irradiation
DiffHor	Horizontal diffuse irradiation
T_Amb	Ambient Temperature
GlobInc	Global incident in coll. plane
GlobEff	Effective Global, corr. for IAM and shadings
EArray	Effective energy at the output of the array
E_Grid	Energy injected into grid
PR	Performance Ratio

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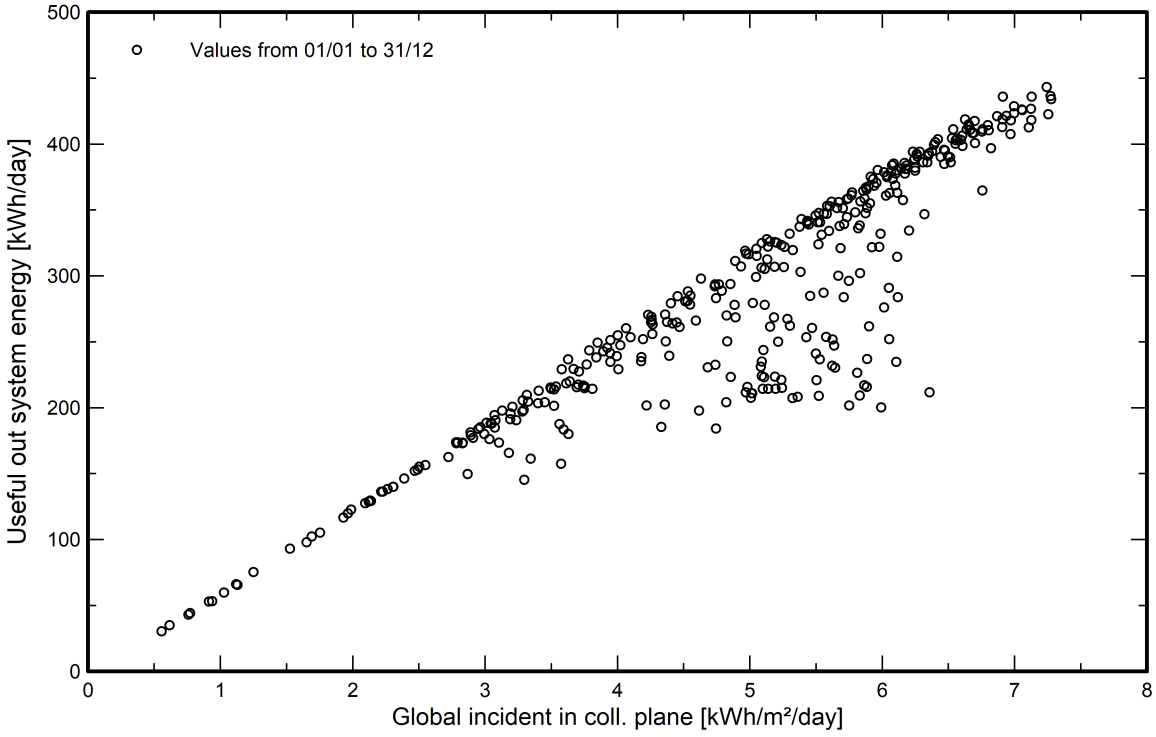
Loss diagram



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Predef. graphs

Daily Input/Output diagram



System Output Power Distribution

